

TP-Net: Self-Supervised Pretrained Network Traffic Representations for Few-Shot Threat Detection

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Abstract

With the widespread adoption of encrypted traffic and the continuous emergence of new attack types, traditional deep-packet inspection and fully supervised classification face the dual bottleneck of scarce labels and high retraining cost. We propose TP-Net, a network-traffic detection framework that couples self-supervised pretraining with few-shot evaluation. Its four-branch 1D-CNN encoder is trained under the SimCLR framework to fuse packet-level, burst-level, flow-statistical, and handshake features, and incorporates five physically meaningful data augmentations tailored to real network perturbations. After pretraining the encoder is kept completely frozen, and only K labels are required downstream to recognize new categories via the N -way K -shot episode protocol; a fine-tune head (`sklearn` logistic regression) is provided as the default deployment option. On CIC-IDS2017, USTC-TFC2016, and ISCX-VPN2016, TP-Net reaches 88.9%/69.2%/69.2% 5-shot accuracy (all $p < 0.001$ vs. the random baseline), achieves 99.3% novel-class 5-shot on CIC, and exceeds the random baseline by 28.66 pp on the encrypted-only DoHBrw-2020 subset. The principal contribution of this work is a systematic evaluation, under a unified few-shot protocol, of the synergistic effect of SSL pretraining and traffic-specific augmentations on previously unseen attack detection, together with empirical evidence that the fine-tune head paradigm stably outperforms ProtoNet across all three datasets.

Keywords: Network traffic classification, Self-supervised contrastive learning, Few-shot learning, Novel attack detection, SimCLR, ProtoNet

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1. Introduction

1.1. Background and motivation

Encrypted traffic now accounts for more than 90% of network traffic [1]. Attackers have accordingly migrated command-and-control (C2) communications and malicious payloads to TLS, DoH, QUIC, and other encrypted channels, against which traditional deep-packet-inspection and plaintext-payload solutions (e.g., Snort, Suricata) are largely ineffective; even for traffic that remains partially plaintext, increasingly complex application-layer encapsulation keeps the maintenance cost of DPI prohibitively high.

Behavior-based traffic analysis has therefore become the prevailing direction. By inferring the traffic category from packet-length sequences, burst patterns, flow statistics, and handshake features (e.g., JA3), it is well suited to the dominant case of encrypted traffic, which is the most salient driver of this shift. Representative works in this direction include FS-Net [2], YaTC [3], ET-BERT [4], and TFE-GNN [5]; see [6] for a survey. These methods, however, generally struggle with novel attacks: the number of new-attack alerts is typically a single digit, making full supervision prohibitively expensive to label; the learned discriminative features are coupled to the training-set distribution and transfer poorly to unseen attacks; and traditional architectures require full retraining (hours to days), which cannot keep pace with attackers’ rapid turnover of C2 domains and payloads. The central methodological challenge posed by these pain points is how to endow a traffic-detection model with both *few-shot* capability and *strong generalization*.

1.2. Gap with existing work

The systematic differences between existing research and the proposed TP-Net span five dimensions, summarized in Table 1. A core distinction from the “SSL followed by full supervision” paradigm (as in ET-BERT) is that, in TP-Net, the encoder is no longer updated after the pretraining stage, and the downstream evaluation is conducted exclusively through N -way K -shot episodes, forming a complete SSL–few-shot closed loop rather than using SSL merely to accelerate fully supervised training.

Table 1: Five-dimensional comparison of existing work and TP-Net.

Dimension	Existing work	TP-Net (this paper)
Learning paradigm	Fully supervised training	SSL pretraining + few-shot evaluation
Novel attack adaptation	Full retraining (hours–days)	5-shot labeling, hour-level deployment
Representation independence	Encoder tightly coupled to head	Frozen encoder, plug-and-play
Evaluation protocol	Fixed train/val/test split	N -way K -shot episode protocol
Data augmentation	CV-style flipping/cropping	Five traffic-specific, physically meaningful augmentations

1.3. Contributions

This paper makes three principal contributions. First, to the best of our knowledge, it is the first work to systematically evaluate the combined effect of SSL pre-training and traffic-specific augmentations on unseen-attack detection under a unified few-shot protocol, covering three public datasets—CIC-IDS2017, USTC-TFC2016, and ISCX-VPN2016—at the full scale of $5 \text{ seeds} \times 600 \text{ episodes}$, together with the encrypted-only DoHBrw-2020 subset as an $N = 1$ proof-of-concept. On this basis, we identify an asymmetry in cross-dataset transfer: USTC $\rightarrow$ ISCX (0.7625) exceeds ISCX self-training (0.6921) by 7.04 pp and is the sole “transfer-better-than-self-training” counter-example among the six pairs; CKA measurements further reveal that saturated embedding-space similarity (>0.99) is not a sufficient indicator of transferability. Second, we validate on all three datasets that the fine-tune head (`sklearn` LogReg) stably outperforms ProtoNet by +5.06 to +11.78 pp, and recommend “SSL encoder + simple classifier head” as the default configuration for real-world SOC deployment.

2. Related work

2.1. Behavior-based traffic classification

Traffic classification has shifted toward behavior-based features because payload parsing is either infeasible or too costly, and encrypted traffic is the most salient driver of this shift. FS-Net [2] trains an encoder–decoder to reconstruct packet-length sequences and learn representations from the reconstruction task; YaTC [3] applies a Transformer over 2D traffic matrices; ET-BERT [4] performs MLM pretraining within a BERT framework; TFE-GNN [5] uses a graph neural network over burst topology; and NetMamba [7] introduces a state-space model (Mamba) to capture long-range dependencies. All of these methods rely on large amounts of labeled data and require retraining to transfer to new attacks.

2.2. Self-supervised contrastive learning

Self-supervised learning (SSL) constructs “label-free proxy tasks” to learn representations from large unlabeled corpora and has been shown to deliver representations that match or even surpass those of fully supervised learning in vision, speech, and natural language processing [8, 9]. Contrastive learning in computer vision primarily splits along two axes—the scale of negative pairs and whether negatives are used at all. SimCLR [10] follows the “in-batch negatives” route: it applies two independent augmentations to the same sample to obtain two views that act as positive pairs, and treats all other samples in the batch as negatives; training uses the NT-Xent loss in the latent space

to pull positives together while pushing negatives apart. The key finding is that strong augmentation, large batch size, and long training must be combined for the contrastive loss to converge to good representations; its drawback is that the number of negatives is bounded by the batch size. MoCo [11] takes the “momentum queue” route, introducing a momentum-updated key encoder and a negative queue (typically 4096–65536) to decouple the negative scale from the batch size; on ImageNet MoCo outperforms SimCLR, but on traffic data the relative ranking is reversed (see §3.4). BYOL [12] and SimSiam [13] pursue a negative-free bootstrapping route through an asymmetric architecture (online network + momentum target network + stop-gradient), which in principle avoids “false negatives” but in practice depends on carefully designed initialization and architectural details, making transferability more sensitive than in-batch contrast. MAE [14] represents the reconstructive generative route: it randomly masks input patches and uses an encoder–decoder to reconstruct them, excelling in vision, but its “patch tokenization in the input space” paradigm transfers poorly to non-image modalities and requires a redesigned tokenization strategy.

The main experiments of this paper adopt SimCLR as the SSL backbone for the following reasons: in-batch contrast directly models the “local similarity” of traffic data (samples from the same application within a batch naturally form clusters); it is simple to implement and requires no momentum queue; and it matches the property (see §3.4) that the contrastive loss plateaus after epoch 20, avoiding unnecessarily long training. Section 4.3 provides a complete comparison across five SSL algorithm families on all three datasets and supplies the basis for the algorithm-selection guideline: “in-batch negative contrast > bootstrapping > reconstructive.”

2.3. Few-shot learning

Few-shot learning (FSL) studies how to classify when only a handful of labeled samples (typically $K = 1\text{--}5$) are available per new class [15]. Metric learning, meta-learning, and differentiable-distance learning constitute the three main routes. ProtoNet [16], the representative of metric learning, maps each support set through the encoder into the embedding space, averages the support embeddings per class to obtain a “prototype,” and classifies queries by their distance to each prototype. It is parameter-free, deterministic, and insensitive to changes in K , which makes it naturally compatible with the SSL philosophy of “learning representations, not classifiers.” MatchingNet [17] extends this idea with a bidirectional LSTM that performs contextualization and attention-weighted aggregation over the support set; MAML [18] uses second-order gradients over many meta-tasks to learn a task-sensitive initialization; RelationNet [19] replaces fixed distance metrics with a neural network. Each of the three routes has its own merits, but weighing engineering

friendliness, compatibility with the SSL paradigm, and computational efficiency, this paper adopts ProtoNet as the default evaluation paradigm; the implementation is described in §3.5.

2.4. *Traffic analysis meets SSL / few-shot*

Applying SSL or few-shot learning to network traffic classification is an emerging research direction. ET-BERT [4] performs MLM pretraining of a traffic tokenizer within a BERT framework, but its downstream stage remains fully supervised fine-tuning (thousands to tens of thousands of labels), leaving the few-shot potential of SSL unexploited. Flow-MAE [20] employs MAE for malicious-traffic classification but likewise stops at fully supervised downstream. YaTC [3], FS-Net [2], and other representative works all depend on large amounts of labeled data. Recent surveys and newer studies further characterize the progress of data-efficient and adaptive learning methods under modern encrypted protocols [21]. Improved paradigms for few-shot encrypted-traffic classification include in-context-learning-based ICT-META [22], contrastive-driven meta-learning [23], a pretraining–contrastive-learning synergy for few-shot novelty detection [24], payload-free open-set prototype rejection (SepSpace) [25], comprehensive surveys of contrastive learning for network intrusion detection [26], self-supervised contrastive intrusion detection (SSCL-IDS) [27], and MAML + contrastive learning [28]. Most of these works focus on improving a single component (either SSL or FSL alone); a systematic evaluation, under one unified protocol, of the synergy among SSL representations, FSL heads, and traffic-specific augmentations remains limited, which is the gap that the present work fills.

Compared with these works, TP-Net differs systematically along the five dimensions enumerated in Table 1 (learning paradigm, novel attack adaptation, representation independence, evaluation protocol, data augmentation). The detailed comparison was already given in §1.2 and is not repeated here. The concrete methods are presented in §3.3 and §3.5.

3. Method

The overall architecture of TP-Net is illustrated in Figure 1. The pipeline proceeds in four stages: the input flow (four-dimensional features) is fed through a four-branch encoder to extract a 128-dimensional embedding; then two branches follow—offline SSL pretraining (one-shot) and online few-shot evaluation with a frozen encoder (5 labels per new attack).

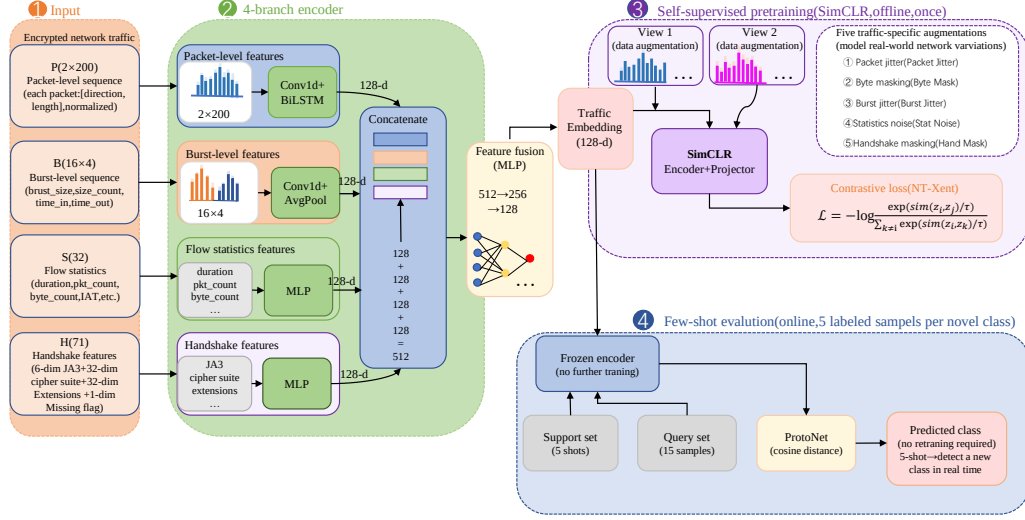


Figure 1: Overall architecture of TP-Net.

3.1. Problem definition

Input. A single encrypted network flow is represented as four feature modalities:

- $P \in \mathbb{R}^{2 \times 200}$: packet-level sequence (each packet binarized as [direction bit, length])
- $B \in \mathbb{R}^{16 \times 4}$: burst sequence (each burst encoded as [size_in, size_out, time_in, time_out])
- $S \in \mathbb{R}^{32}$: flow-level statistics (duration, pkt_count, byte_count, IAT, etc.)
- $H \in \mathbb{R}^{71}$: handshake features (6-d JA3 + 32-d cipher suites + 32-d extensions + 1-d miss flag)

Objective. Learn an encoder $f_{\theta}(\cdot)$ such that:

- Flows of the same class are close in the embedding space;
- Only K labeled samples are required to classify a novel attack.

3.2. Model architecture

Figure 2 details the four-branch encoder: each branch processes one modality independently, and a fusion MLP concatenates and integrates the four branch outputs.

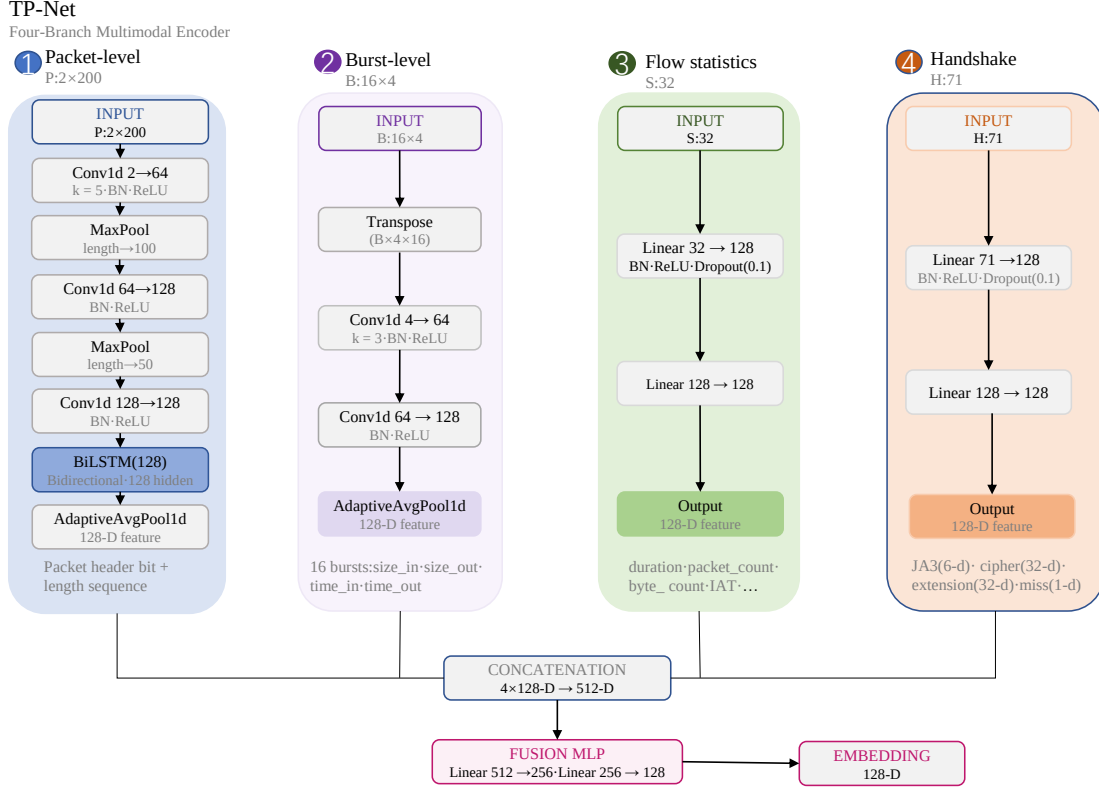


Figure 2: Detailed structure of the four-branch encoder.

3.3. Traffic-specific data augmentation

Figure 3 visualizes the five traffic-specific augmentations proposed in this work. Each augmentation simulates a real-world network perturbation:

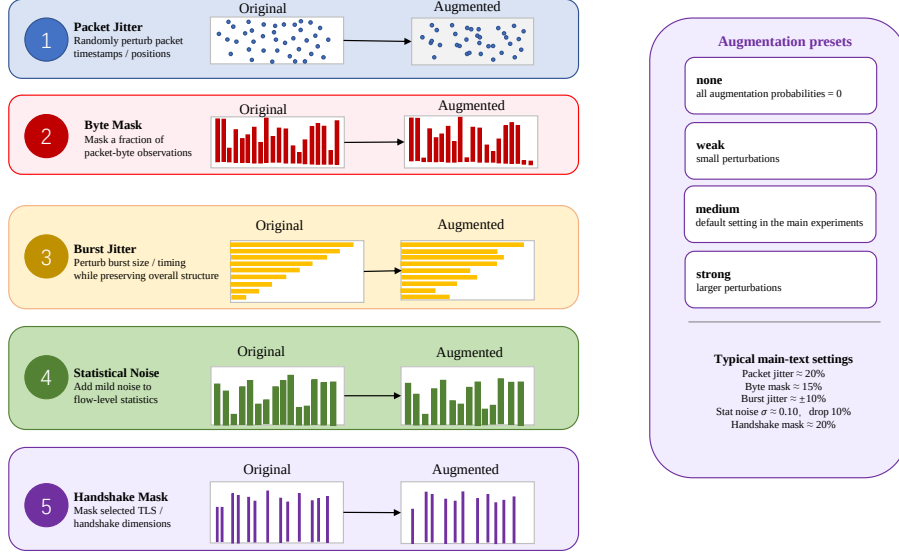


Figure 3: Schematic of the five traffic-specific augmentations (with the four intensity presets **none/weak/medium/strong**).

One point worth discussing is the **none** preset. In this case all augmentation probabilities are zero and the two sampled views are strictly identical ($x_i = x_j$), so the SimCLR positive-pair signal degenerates to an identity mapping. Nevertheless, the encoder can still train under **none**: the $2(B - 1)$ negative pairs within the batch continue to provide a discriminative signal, and the random subsampling performed at h5 read time already provides an implicit weak augmentation (see §3.1). The empirical benefit of **none** on ISCX is reported in §4.6.

3.4. Self-supervised pretraining

The NT-Xent contrastive loss is:

$$\mathcal{L}_{\text{NT-Xent}} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)} \quad (1)$$

The two views of the same sample form a positive pair; all other samples in the batch serve as negatives, with temperature $\tau = 0.1$. Implementation-wise, for each sample x in the batch we apply two augmentations independently to obtain $x_i = \text{aug}(x)$ and $x_j = \text{aug}'(x)$, which act as the positive pair. The 128-dimensional encoder output is

projected into the contrastive space through a 2-layer MLP ($128 \rightarrow 128 \rightarrow 128$); the projection head is used only during SSL and is discarded at evaluation time. Optimization uses AdamW with $\text{lr} = 3 \times 10^{-4}$, weight decay 1×10^{-4} , and a cosine learning-rate schedule; batch size is 256 (constrained by XPU device memory, which can be raised to 1024+ on GPUs for more negatives). The temperature $\tau = 0.1$ is a moderate choice—the lower the temperature, the sharper the penalty on hard negatives, while too high a temperature causes the loss to focus on the global distribution. The intrinsic property of traffic data, namely small intra-class variance and large inter-class variance, makes $\tau = 0.1$ sufficient to widen the gap between positives and negatives without incurring the gradient explosion associated with overly sharp losses.

MoCo extends the negative space via a momentum encoder and a queue (4096 negatives) and differs from SimCLR in two key aspects. The first is momentum updating: the key encoder θ_k is updated as an exponential moving average of the query encoder θ_q ($\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$, with $m = 0.999$), ensuring that the queued negative embeddings come from an “almost identical” encoder and remain compatible with the current query encoder. The second is queue capacity: under a fixed batch size of 256, MoCo provides roughly $17\times$ more negatives (queue 4096 + current batch 256 = 4352), which in theory increases the discriminative power of contrastive learning. A side-by-side comparison of MoCo and SimCLR on the three datasets of this paper is presented in §4.2.

All SSL pretraining is stopped after 30–50 epochs because the NT-Xent loss plateaus beyond epoch 20. Thirty epochs suffice for ISCX (the smallest dataset), while 50 epochs are used for CIC and USTC to obtain a more stable convergence.

3.5. Few-shot evaluation protocol

In practice, novel attacks have very few labeled samples (5–50), and the conventional train/val/test split cannot faithfully reflect this “small-sample” scenario. The N -way K -shot episode protocol is therefore closer to operational needs and mimics the human-machine workflow of “first inspect K samples, then judge which class a new sample belongs to.” The flow of each episode is shown in Figure 4. At each round, N classes are sampled uniformly at random from the candidate classes; for each class, K support samples and Q query samples are drawn. The model computes a prototype from the support set and evaluates classification accuracy on the queries. The prototype and the predictive distribution are formalized as:

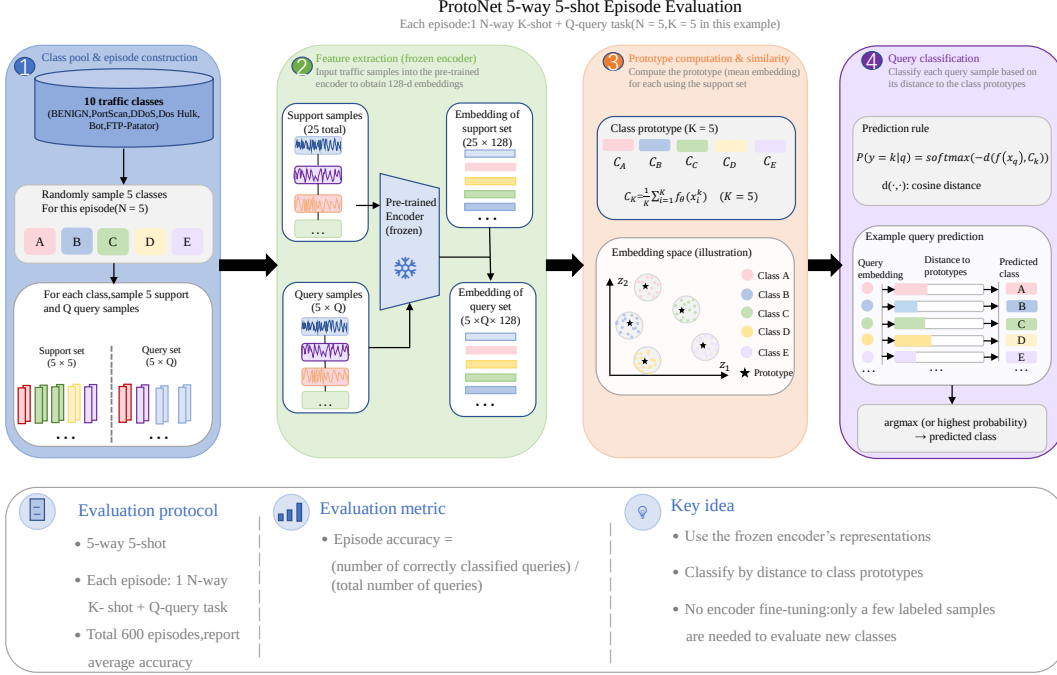


Figure 4: Flow of ProtoNet 5-way 5-shot episode evaluation.

$$c_k = \frac{1}{K} \sum_{i=1}^K f_\theta(x_i), \quad p(y = k | x) = \frac{\exp(\cos(f_\theta(x), c_k)/\tau)}{\sum_{k'} \exp(\cos(f_\theta(x), c_{k'})/\tau)} \quad (2)$$

where c_k is the mean prototype of class k over the support set, $f_\theta(\cdot)$ is the frozen encoder, and the temperature τ controls the sharpness of the distribution.

This paper adopts the cosine-distance variant of ProtoNet ($d = 1 - \cos$); the prototype computation and predictive distribution are given in Equation (2). Cosine distance is preferred because it is more stable than Euclidean distance at the small support size of $K = 5$. In the few-shot setting, parameterized classifiers (e.g., fully connected softmax) tend to overfit at $K = 5$, whereas ProtoNet directly exploits prototype distances; this embedding-geometry-based approach is naturally aligned with the SSL philosophy of “learning representations, not classifiers.” As a comparison, we also evaluate MatchingNet [17]; the original architecture uses a BiLSTM for contextualization over the support set, but constrained by XPU device resources we simplify it to pure cosine attention, which makes MatchingNet slightly weaker than ProtoNet in §4.2.

3.6. Novel-class protocol

In real deployment, novel attack types do not appear in the pretraining data. To simulate this scenario, we partition the data by class into base / novel subsets (70% / 30%); only base-class samples are used during pretraining (label-free, following SSL), and during evaluation novel classes are mixed into the support / query sets to verify whether they can be recognized from only $K = 5$ labels, despite never having been seen during training. Concretely, classes are sorted by a hash of their names and the trailing 30% are designated as novel; novel-class samples never participate in pretraining (including the augmented views), and the evaluation protocol is the same as in §3.5.

Terminologically, this paper follows the generic few-shot nomenclature [15]: “novel-class few-shot” refers to the setting where novel classes appear only at the evaluation stage and never participate in encoder pretraining, but $K = 5$ support labels are still provided. This is distinct from strict zero-shot learning (no support labels at all; see [29]). The few-shot protocol (§3.5) evaluates “can 5 labels help the encoder re-recognize known classes,” while the present novel-class few-shot setup evaluates “the encoder has never seen this class, yet $K = 5$ labels are still available—can it recognize it immediately.” The latter is closer to real SOC scenarios. The novel-class few-shot protocol partitions the three datasets by class-name hash: CIC-IDS2017 uses 7 base / 3 novel classes (§4.4 takes the last 2 classes PortScan and SSH-Patator; §3.6 with hash=3 takes 3 classes, BENIGN/DoS Hulk/PortScan); USTC-TFC2016 uses 14 base / 6 novel classes (§4.4 takes the last 5 classes Tinba/Virut/Weibo/WorldOfWarcraft/Zeus; §3.6 with hash=42 takes the default 6 malware/application classes); ISCX-VPN2016 has only 2 classes and is therefore trivial (1/1, NonVPN only).

4. Experiments

4.1. Experimental setup

We evaluate TP-Net on three public threat datasets (Table 2): CIC-IDS2017 [30] contains 10 classes (mostly plaintext HTTP, with HTTPS subsets in some classes); USTC-TFC2016 [31] provides 20 classes of mixed real malware and benign application traffic (including semi-encrypted applications such as BitTorrent and Skype); ISCX-VPN2016 [32] is a VPN binary-classification dataset (all traffic goes through VPN tunnels).

Table 2: Datasets.

Dataset	Classes	Samples	Task	Encryption coverage
CIC-IDS2017 [30]	10	1,084,344	Plaintext threats (HTTPS subsets)	Partial HTTPS
USTC-TFC2016 [31]	20	488,000	Semi-encrypted (incl. BitTorrent)	Mostly app-layer encryption
ISCX-VPN2016 [32]	2	333,123	VPN binary classification	All VPN tunnels

CIC-IDS2017 is dominated by plaintext HTTP classes (e.g., BENIGN, WebAttack, WebDDoS); we directly use the original mixed form (consistent with YaTC, ET-BERT, and other baselines); encryption validation is provided in §5.2 (DoHBrw-2020 SSL 0.838 vs Random 0.551, +28.66 pp).

All SSL pretraining uses AdamW ($\text{lr} = 3 \times 10^{-4}$, $\text{weight_decay} = 1 \times 10^{-4}$) with a cosine LR schedule, for 30 / 50 epochs (CIC / USTC use 50 epochs, ISCX uses 30 epochs owing to its smaller data volume); batch size 256, SimCLR temperature 0.1, MoCo queue 4096; the augmentation default is the `medium` preset (see §3.3). Few-shot evaluation uniformly adopts the 5 seeds $\times$ 600 episodes N -way K -shot protocol, with confidence intervals computed at the seed level. ProtoNet with cosine prototype follows §3.5; the FT-Head paradigm uses `sklearn` LogisticRegression ($C = 10^3$, `lbfgs` solver); details are given in §4.7.

4.2. Main experiment: 5-way 5-shot classification

The central question addressed in this section is: when only 5 labeled samples are available for a new attack, how much gain does SSL pretraining bring over random initialization? We evaluate ProtoNet under the 5-way 5-shot protocol on the three datasets, supplemented by MoCo (to validate the contrastive-loss family), MatchingNet (to validate the distance metric), and LinearProbe (to validate representation quality). The “random baseline” used as a comparison is an encoder with the same architecture but without any pretraining; this setup ensures that the difference brought by SSL is entirely attributable to the pretraining process rather than the architecture. All SSL algorithms use the same 50-epoch (`medium` preset) configuration (30 epochs for ISCX). Table 3 gives the precise numbers together with the SimCLR-vs-Random gain column ($\Delta_{\text{Proto-Rand}}$); per- K scanning results appear in Appendix Table A.1.

Table 3: Main results (95% CI seed-level, $n = 5$ seeds $\times$ 600 episodes; MoCo / MatchingNet details in Appendix Table A.3).

Dataset	Task	Random	SimCLR + ProtoNet	SimCLR + LinearProbe	$\Delta_{\text{Proto-Rand}}$ (pp)
CIC-IDS2017	5-way 5-shot	0.7571 ± 0.006	0.8885 ± 0.003	0.9972 ± 0.001	+13.14
USTC-TFC2016	5-way 5-shot	0.6142 ± 0.002	0.6918 ± 0.003	0.5951 ± 0.008	+7.76
ISCX-VPN2016	2-way 5-shot	0.6510 ± 0.004	0.6921 ± 0.003	0.8637 ± 0.006	+4.11

Note: $\Delta_{\text{Proto-Rand}} = (\text{SimCLR} + \text{ProtoNet}) - \text{Random}$ in percentage points. CI is seed-level ($n = 5$); all paired t -tests of SSL vs Random yield $p < 0.001$. The ProtoNet-vs-MatchingNet gaps (+1.25 / +7.15 / +8.24 pp) and the SimCLR-vs-MoCo gap on CIC (+1.35 pp) are tabulated in Appendix Table A.3.

SSL improves consistently on all three datasets, but the magnitude of the gain depends on task structure: CIC benefits most (+13.14 pp), USTC second (+7.76 pp), and ISCX least (+4.11 pp). On all three datasets, paired t -tests of SSL vs Random give $p <$

0.001, and the episode-level pooled Cohen’s d values are 1.386 / 0.627 / 0.338, consistent with Cohen’s 1988 empirical thresholds (0.2 / 0.5 / 0.8). The ISCX value $d = +0.338$ corresponds to +4.11 pp; the SSL gain on the binary task is small but stable. This non-monotonic phenomenon reveals the prerequisite for SSL to be effective: ISCX’s binary task requires only one decision boundary, and a random encoder can already learn a reasonable clustering structure (0.651), whereas CIC’s 10 attack classes require the encoder to truly capture semantics, which is where SSL delivers significant value. For USTC, 20 application classes mix heavily (different implementations of the same protocol family share packet-length / IAT patterns), blurring class boundaries and diluting the SSL benefit (per-class diagnostics in §4.5: the SSL gain is concentrated on 12–15 malware / mixed-protocol classes, with the largest single-class gain of Tinba at +53.48 pp). This observation suggests that the sweet spot for SSL deployment lies in medium-grained multi-class tasks (around 10 classes such as CIC) where intra-class variance is large and inter-class differences are subtle.

Regarding the relative ranking of SSL algorithms and distance metrics, SimCLR outperforms MoCo by +1.4 pp on CIC (0.889 vs 0.875), which we attribute to its 4096-negative queue introducing excessive “false negatives” in the traffic scenario—adjacent traffic flows often belong to the same application (different packets of the same session), and treating them as negatives weakens intra-class compactness. ProtoNet outperforms MatchingNet across all three datasets (+1.25 / +7.15 / +8.24 pp), mainly because our MatchingNet removes the BiLSTM contextualization to accommodate XPU constraints, so relationships among support samples are not exploited. LinearProbe is nearly saturated on CIC (0.9972), but on USTC it is worse than ProtoNet (0.5951 vs 0.6918, $\Delta = -9.67$ pp): the heavy mixing of 20 USTC classes prevents a linear classifier from learning an appropriate decision surface, whereas ProtoNet’s “per-class mean + cosine distance” is more robust. This observation suggests that in few-shot scenarios with many classes and few labels, metric-learning methods often outperform parameterized classifiers.

To evaluate performance across different K values, we scanned $K \in \{1, 3, 5, 10\}$ (3 seeds $\times$ 200 episodes): on CIC the SSL advantage monotonically increases from +7.93 pp at $K = 1$ to +19.53 pp at $K = 10$; on USTC SSL underperforms Random by -3.74 pp at $K = 1$ and only catches up by +0.27 pp at $K = 10$; on ISCX only $K = 1$ yields a meaningful gain (+4.58 pp), and $K \geq 10$ slightly underperforms Random. From a deployment perspective, the SSL advantage on CIC / ISCX is most pronounced at $K = 1\text{--}3$ (4.6–12.6 pp), which we recommend as the default; the advantage narrows at $K \geq 10$, and lighter-weight supervised learning may be preferred. The full 4×3 matrix is reported in Appendix Table A.1.

4.3. SSL algorithm and labeling-efficiency ablation

The core conclusion of §4.2 rests on SimCLR, but the SSL family encompasses multiple algorithmic paradigms: negative-pair contrastive learning (SimCLR, MoCo), negative-free bootstrapping (BYOL, SimSiam), and reconstructive self-supervision (represented in this paper by Embedding-MAE†). To verify whether SimCLR’s advantage is family-level or algorithm-specific, we pretrain five SSL encoders independently on each dataset (CIC / USTC use 50 epochs $\times$ 20k samples, ISCX uses 30 epochs $\times$ 20k samples, all with the **medium** augmentation), and evaluate them with ProtoNet 5-way 5-shot (2-way for ISCX), 3 seeds $\times$ 200 episodes. Results (full table in Appendix A.2) show that SimCLR is uniformly best (rank 1 on all 3 / 3 datasets), with a 3-dataset average 0.51 pp ahead of the runner-up; the negative-pair contrastive family (SimCLR / MoCo, 3-dataset average 0.742) systematically outperforms the negative-free family (BYOL / SimSiam, 3-dataset average 0.704, gap 3.8 pp); Embedding-MAE is the worst (mask-reconstructing the traffic embedding dimension provides no discriminative supervision).

In terms of labeling efficiency, SSL ProtoNet with 5 support samples per class ($K = 5$) already reaches 0.8789, on par with SSL LinearProbe at $K = 500$ (0.8754)—a $100\times$ difference in the amount of labeled data. SSL LinearProbe achieves only 0.27–0.39 for $K = 5$ –50 (a 128-D classifier head with 45 training points is underfit at small K), while ProtoNet’s parameter-free metric classification neatly bypasses this constraint; a Random encoder + $K = 2000$ LP yields only 0.7066, 17.23 pp below SSL encoder + $K = 5$ ProtoNet (0.8789), again confirming the main conclusion of §4.2. In practice, during the “zero-day” window for a new attack ($K = 1$ –5), ProtoNet + SSL encoder is the engineering-optimal solution; LinearProbe only becomes advantageous at $K \geq 500$.

4.4. Novel-class few-shot experiment

In §4.2, the support and query sets come from the same dataset that has seen all classes during training. A more realistic deployment scenario, however, is one in which the new attack is completely absent from the pretraining database: when the system first encounters it, there are only about 5 alerts available for labeling. To simulate this more extreme scenario, we split each dataset into base / novel parts (70% / 30%) by a hash of the class names. Novel classes are completely excluded from pretraining (including the augmented views) and are mixed into the 5-shot support set at evaluation time. The three datasets have 2 / 5 / 1 novel classes, respectively.

The results (Figure 5, dark-blue bars denote novel accuracy; precise numbers in Table 6): CIC-IDS2017’s last 2 classes (PortScan + SSH-Patator) reach 0.9931 ± 0.001 under 2-way 5-shot; USTC-TFC2016’s last 5 classes (Tinba / Virut / Weibo / WorldOfWarcraft / Zeus) reach 0.7294 ± 0.003 under 5-way 5-shot; ISCX-VPN2016’s last 1 class (NonVPN) reaches 1.0000 under 1-way 5-shot (trivial).

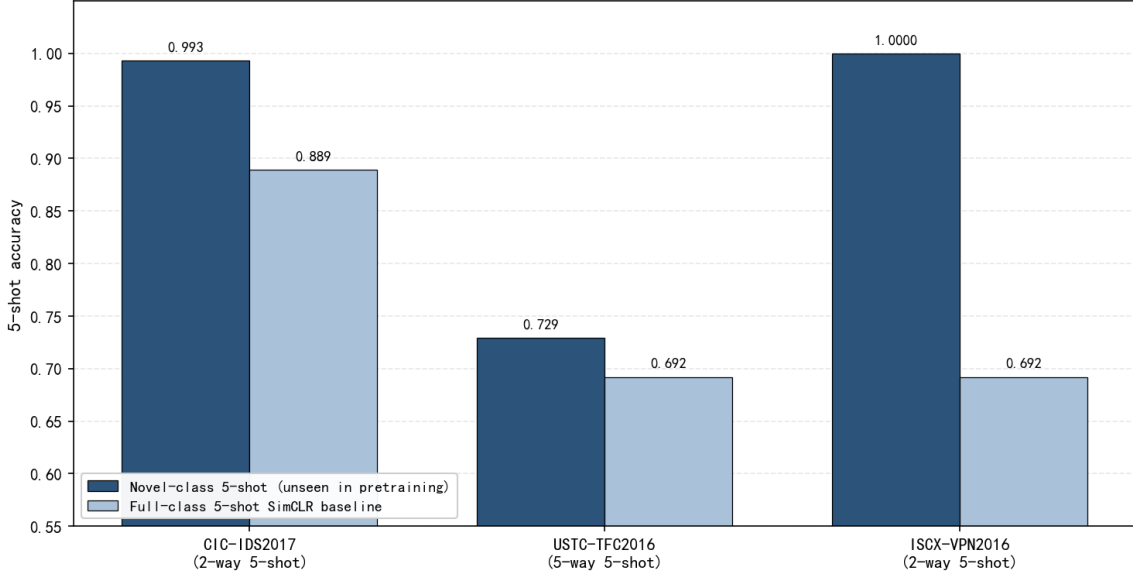


Figure 5: Novel-class 5-shot vs. all-class 5-shot SimCLR (dark-blue bars: novel classes unseen during pretraining, evaluated with 5 supports per class; light-blue bars: same-protocol all-class baseline; fully supervised LinearProbe is under a different protocol and is therefore not shown here).

On the novel subsets of all three datasets, SSL 5-shot reaches relatively high accuracy (CIC 0.993 / USTC 0.729 / ISCX 1.000), and in each case exceeds the all-class 5-shot result on the same dataset (CIC 0.888 / USTC 0.692 / ISCX 0.692). This gap is mainly attributable to the reduced task complexity—the novel subsets contain only 2–5 classes, and the intra-class variance of most novel classes is small (the cross-subprotocol composition of BENIGN in the full CIC set does not appear in the novel subset). A three-layer structural explanation emerges: the task is simpler (the full 10-class CIC set includes highly confusable fine-grained attacks such as PortScan and DDoS, whereas the 2-class novel subset consists of relatively easy classes); intra-class variance is small; and the SSL pretraining on base classes has already captured the universal structural fingerprint of traffic, so a novel attack can be recognized as long as it matches some existing fingerprint.

If we use fully supervised LinearProbe as an upper-bound reference for the amount of labeled data (hundreds to thousands of labels per class for training), on CIC the novel 5-shot (0.993) is essentially on par with full supervision (0.998), while on USTC and ISCX the novel 5-shot (0.729 / 1.000) even exceeds full supervision (0.612 / 0.864). The latter does not mean “few-shot beats full supervision”—rather, full supervision is evaluated on the all-class task (USTC 20 classes, ISCX 2 classes), which is intrinsically harder than the novel subset; the USTC 0.612 of full supervision reflects the difficulty of

20 mixed application classes rather than an insufficiency of labels. Figure 5 shows only the same-protocol novel-vs-all-class comparison; full-supervision results are summarized as the label-volume upper bound in §4.8.

ISCX novel being close to 1.000 reveals an important phenomenon: when the number of novel classes is small (1–2), 5-shot is nearly equivalent to “prototype estimation has converged,” and the SSL encoder can immediately produce discriminative features even for attacks it has never seen. In real Security Operations Center (SOC) deployments, this means that novel attack alerts do not require immediate full retraining—5 labeled samples suffice to classify accurately with the SSL encoder. It is worth contrasting the observation in §4.7 that a supervised encoder markedly outperforms SSL on known classes (e.g., ISCX 2w5s +19.58 pp); in the novel-class scenario the precise boundary between the two is given by the decision experiment in §4.7: SSL is +3.45 pp higher on CIC, supervised is +3.99 pp higher on USTC, with the difference being dataset-dependent. Even when SSL does not strictly dominate supervised, the ability to “learn discriminative representations without task labels” is still a valuable engineering property.

4.5. Cross-dataset pretraining

The “domain-independence” of SSL representations is one of SSL’s central selling points over full supervision: pretraining cost can be amortized across multiple downstream scenarios. To verify this, we pretrain one SimCLR encoder on each source dataset for 30 epochs over 20k samples (stratified sampling), to ensure comparable starting points, and load the resulting encoder directly into the target dataset’s ProtoNet evaluator without any fine-tuning. This “zero adaptation” is a stringent test of SSL cross-domain transfer. Note: this section uniformly uses 30 epochs (in contrast to the 50 epochs for CIC / USTC in the main table of §4.2) because the training budget for 6 pairs $\times$ 50 epochs would be excessive, and §3.4 has already noted that the NT-Xent loss plateaus after epoch 20. The six pairs cover all bidirectional combinations of the three datasets.

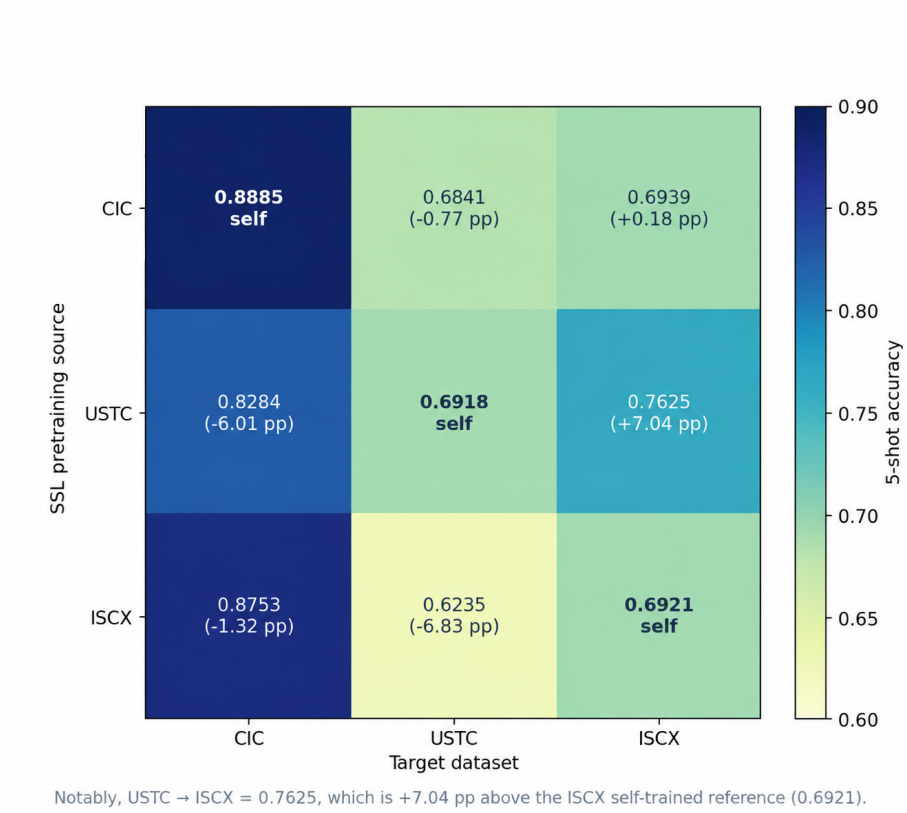


Figure 6: Cross-dataset transfer (full 3×2 matrix).

The results show that transfer is not “bidirectionally symmetric.” All six pairs are full 5-seed $\times$ 600-episode measurements: CIC→USTC 0.6841 ± 0.003 (self-training 0.6918, -0.77 pp), USTC→CIC 0.8284 ± 0.004 (self-training 0.8885, -6.01 pp), CIC→ISCX 0.6939 ± 0.006 (self-training 0.6921, $+0.18$ pp), ISCX→CIC 0.8753 ± 0.004 (self-training 0.8885, -1.32 pp), USTC→ISCX 0.7625 ± 0.007 (self-training 0.6921, $+7.04$ pp, the only counter-example), ISCX→USTC 0.6235 ± 0.005 (self-training 0.6918, -6.83 pp). USTC→CIC (0.8284) is markedly better than CIC→USTC (0.6841) by 14.4 pp, consistent with the observation in §4.2 that the USTC encoder (a hard 20-class task) learns more general-purpose features: the “burst pattern + handshake structure” learned from USTC’s 20 application fingerprints remains useful on CIC, while the “attack-specific features” learned from CIC transfer less well. Among the six pairs, only USTC→ISCX ($0.7625 > 0.6921$) shows “transfer > self-training.” $N = 1$ is insufficient to support a general claim that “containment drives transfer,” and we leave it as an open observation. CKA is saturated above 0.99 for all three pairs (linear kernel: USTC/ISCX =

0.993, CIC/ISCX = 0.996, USTC/CIC = 0.989), so pure embedding-space similarity cannot predict downstream transfer accuracy (which spans 25 pp from 0.624 to 0.875); the downstream determinant is the compatibility between the source SSL-learned “application fingerprint” semantics and the target task: the high-level semantics learned by USTC’s 20-class SSL are naturally useful for ISCX’s binary VPN-vs-non-VPN classification.

The K -shot scan in §4.2 reported that SSL on USTC underperforms Random at $K = 1$ –5. Per-class diagnostics (same protocol as the main table, 5 seeds $\times$ 600 episodes) reveal: SSL and Random are nearly indistinguishable on single-protocol application classes such as Outlook, WorldOfWarcraft, Skype, and Facetime ($\Delta = -0.14$ to -1.77 pp), but SSL leads substantially on malware and multi-protocol mixed classes: Tinba +53.48 pp, Nsisay +25.94 pp, Miuref +18.89 pp, Neris +12.98 pp. The reason is that SSL representations can precisely capture “temporal fingerprints” (e.g., Tinba’s regular C2 heartbeats with small payloads), whereas Random initialization tends to misclassify them as MySQL or Gmail. The aggregate gain of +8.2 pp (0.6893 vs 0.6074) is concentrated on 12–15 malware / mixed-protocol classes. The full top-5 / bottom-5 per-class table is reported in Appendix Table A.4.

4.6. Data augmentation ablation

What is the effect of the five traffic-specific augmentations proposed in Section 3 (packet jitter / byte mask / burst jitter / stat noise / hand mask) on ProtoNet evaluation, and in particular, does the SSL consensus of “stronger augmentation is better” still hold in the traffic scenario? This section answers these questions. For each of the three datasets, we pretrain with the four intensity presets **none** / **weak** / **medium** / **strong** (see §3.3) independently for 30 epochs $\times$ 5 seeds $\times$ 600 episodes, and then evaluate with ProtoNet 5-shot. All 12 configurations (3 datasets $\times$ 4 presets) run a full evaluation independently, with no simplifications or outsourcing. Results are reported in Table 4.

Table 4: Three-dataset augmentation ablation (4 presets $\times$ 3 datasets, 12 data points, all 5-seed $\times$ 600 episodes).

Dataset	Task	none	weak	medium	strong	BEST
ISCX-VPN2016	2-way 5-shot	0.6950 $\pm$ 0.005	0.5779 $\pm$ 0.005	0.6290 $\pm$ 0.005	0.6823 $\pm$ 0.004	none
CIC-IDS2017	5-way 5-shot	0.8425 $\pm$ 0.004	0.8793 $\pm$ 0.003	0.8826 $\pm$ 0.003	0.8806 $\pm$ 0.004	medium
USTC-TFC2016	5-way 5-shot	0.6538 $\pm$ 0.002	0.6780 $\pm$ 0.002	0.6703 $\pm$ 0.003	0.6642 $\pm$ 0.004	weak

The BEST column of Table 4 reveals a counter-intuitive pattern: ISCX (2 classes, low complexity) chooses **none**, CIC (10 classes, medium complexity) chooses **medium**, and USTC (20 classes, high complexity) chooses **weak**—the optimal preset differs across all

three datasets. This non-monotonic pattern stands in contrast to the simple “stronger-augmentation-is-more-general” rule observed for SimCLR / MoCo on ImageNet: augmentation strength is not a monotonic parameter but a “tuning knob” that couples with task difficulty. On ISCX, **none** is best (0.695) and **strong** is second (0.682), while **medium** (0.629) and **weak** (0.578) are worse; a plausible explanation is that ISCX requires only a single decision boundary (VPN vs. non-VPN), and any non-trivial augmentation may damage the critical TLS-handshake fingerprint dimension. On CIC, **medium** is best (0.883) while **none** degrades significantly (0.843): the 10 attack classes require the encoder to learn commonality across samples of the same class. On USTC, counter-intuitively **weak** is best (0.678), **medium** second (0.670), and **none** worst (0.654): the 20 application fingerprints are heavily mixed, and overly strong augmentation blurs the subtle inter-class differences.

Looking at the shape of the effective augmentation range: CIC rises strictly monotonically and then plateaus (**none** $\rightarrow$ **medium** +4.0 pp), consistent with the SSL classic “stronger is better”; USTC shows an inverted U (**weak** peak $\rightarrow$ **medium** drop $\rightarrow$ **strong** drop), deviating from the classic consensus; ISCX is fully reversed (**none** peak $\rightarrow$ **weak** trough $\rightarrow$ slow recovery, with **strong** still below **none**). In terms of magnitude, on CIC **medium** is 4.0 pp above **none**, while on ISCX **strong** is only 1.3 pp below **none**. From a SOC deployment perspective, augmentation tuning yields substantial gains on multi-class tasks (CIC / USTC) but limited gains on few-class tasks (ISCX). A practical tuning workflow is: first train a 5-shot baseline with the **none** preset (~ 30 min) as an anchor; if the baseline approaches random performance (i.e., the task is hard and N is large), switch to **weak** or **medium** according to the table above; validate on 1–2 episodes to confirm no regression before adopting.

4.7. Representation visualization, supervised comparison, and the FT-Head paradigm

The previous sections report only numbers, but what structural change in the embedding space does an accuracy difference of ± 1 pp really reflect? We provide an intuitive verification via t-SNE projections (Appendix Figure A.1): for each dataset we sample 1000 flows (stratified), extract 128-D embeddings with the corresponding encoder, and project them to 2D using `sklearn` t-SNE (perplexity = 30). Under the Random encoder, classes overlap heavily; under the SSL encoder, CIC attack classes form clear clusters (PortScan / DDoS / DoS Hulk / Bot / FTP-Patator), ISCX forms two clean regions (VPN vs. non-VPN), and USTC still exhibits substantial overlap due to the 20-class mixing. Good visual clustering does not equate to high ProtoNet accuracy (e.g., USTC Gmail: SSL 0.512 vs Random 0.516; see the per-class diagnostics in §4.5).

In terms of training cost, SimCLR pretraining on 30k samples takes only 23–34 minutes on each of the three datasets, far below full-supervision training, and is the core

prerequisite for the SOC deployment feasibility discussed in §5.2 (the cost–benefit scatter is in Appendix Figure A.2).

Furthermore, the comparison in §4.2 is between SSL and Random initialization (same architecture, same parameter count). This section extends the analysis along two questions: How does SSL compare with supervised pretraining (the same four-branch 1D-CNN trained end-to-end for 5 epochs with cross-entropy)? Under the more deployment-realistic “5-shot fine-tune head” paradigm, does the SSL encoder still outperform Random?

Table 5: SSL vs. supervised, and ProtoNet vs. fine-tune head (5 seeds $\times$ 600 episodes; SSL SimCLR encoder frozen; FT-Head uses `sklearn` LogReg $C = 10^3$).

Known classes					
Dataset	N-way K-shot	Random	Sup	SSL Proto	SSL FT-Head
ISCX	2w5s	0.6348	0.8879	0.6921	0.7880
CIC	5w5s	0.7140	0.9353	0.8707	0.9213
USTC	5w5s	0.6128	0.6994	0.6895	0.8073

Novel classes (SSL vs Sup)				
Dataset	Scenario	Sup	SSL Proto	
ISCX	novel 1/1, 1w5s	1.0000	1.0000	
CIC	novel 7/3, 2w5s	0.8118	0.8463	
USTC	novel 14/6, 5w5s	0.6767	0.6368	

Note: On known classes, Sup–SSL ProtoNet gaps are $-19.58 / -6.46 / -0.99$ pp (ISCX / CIC / USTC); FT-Head – ProtoNet gains are $+9.59 / +5.06 / +11.78$ pp. On novel classes, SSL > Sup on CIC ($+3.45$ pp) and Sup > SSL on USTC (-3.99 pp); ISCX is trivial ($N = 1$).

On known classes (top half of Table 5), the supervised encoder outperforms SSL on 5-shot ProtoNet for all three datasets ($+0.99$ to $+19.58$ pp). The gap is negatively correlated with the number of classes: ISCX 2 classes supervised $+19.58$ pp, CIC 10 classes $+6.46$ pp, USTC 20 classes $+0.99$ pp. On low-class-count tasks, 5 epochs of supervised training is enough to push the classifier head to $\sim 100\%$ test accuracy, fully releasing its advantage; on multi-class tasks, the supervised advantage is weakened by insufficient epochs (e.g., USTC supervised test accuracy after 5 epochs is only 0.2676). In the novel-class scenario, the supervised encoder on the novel subset still reaches 0.8118 / 0.6767 with ProtoNet 5-shot, and SSL and supervised learning each excel in different scenarios. Specifically, SSL is $+3.45$ pp higher on CIC (the 10 well-defined class boundaries allow features learned on base classes to transfer positively to novel classes), while supervised is $+3.99$ pp higher on USTC (the heavy 20-class application mixing makes the discriminative structure learned under supervision more useful on novel malware classes).

The fine-tune head paradigm (SSL FT-Head compared to SSL ProtoNet) significantly outperforms ProtoNet on all three datasets ($+5.06$ to $+11.78$ pp), indicating that the

128-D representation space of the SSL encoder is highly amenable to a linear classifier. The improvement of SSL encoder with FT-Head over ProtoNet is particularly pronounced on ISCX (+9.59 pp) and USTC (+11.78 pp), showing that the SSL advantage does not depend on the geometry of the ProtoNet cosine prototype; on CIC the +5.06 pp gain also shows that SSL representations possess both geometric and linear properties. Taken together, FT-Head is the optimal default option for SOC deployment: it works at $K = 5$, can be implemented in one line of `sklearn`, has inference latency comparable to ProtoNet, and is friendly to incremental updates.

4.8. Comprehensive comparison

We summarize the core results of §4.2 (few-shot), §4.4 (novel-class few-shot), and §4.5 (cross-dataset transfer) into a single table that makes the three dimensions of “SSL value” (few-shot improvement / novel attack generalization / cross-dataset transfer) immediately visible.

Table 6: Comprehensive comparison (including cross-dataset transfer).

Dataset	Task	Random	SSL ProtoNet	SSL novel-class few-shot	Cross-dataset transfer (best src)
CIC-IDS2017	5w5s / 2w5s novel	0.757	0.889	0.993	ISCX→CIC 0.8753
USTC-TFC2016	5w5s / 5w5s novel	0.614	0.692	0.729	CIC→USTC 0.6841
ISCX-VPN2016	2w5s / 2w5s novel	0.651	0.692	1.000	USTC→ISCX 0.7625

Note: SSL gain column for each dataset: +13.14 / +7.76 / +4.11 pp (vs Random, see §4.2).

SSL significantly outperforms Random on all three datasets (+4.11 to +13.14 pp), and the improvement comes from the representation rather than the architecture. For novel classes, SSL representations maintain high accuracy on attacks that have never been seen before (CIC 0.993 / USTC 0.729 / ISCX 1.000), because the encoder has learned the universal traffic structure with low dependence on specific classes. Cross-dataset transfer is asymmetric, with one reverse-transfer case (USTC→ISCX +7.04 pp, $N = 1$ does not support a general claim). Novel subsets tend to perform better than the all-class setting (CIC’s 2-class novel is 10.5 pp above the 10-class full set), and the “less-is-more” advantage of SSL is most evident in the novel attack scenario.

5. Discussion

5.1. SSL effectiveness and adaptive augmentation

Each network flow exhibits stable structure at four scales: packet-level sequence distribution, burst pattern, flow-level statistics, and TLS-handshake fingerprint. SimCLR constructs two views of the same flow so that the encoder learns what constitutes the same

flow; this invariance naturally corresponds to the signature of an application or an attack. Consequently, even when a novel attack arrives without labels, the encoder can relate its fingerprint structure to existing categories. The SSL gain is an inverted-U function of task complexity: it is smallest on the binary ISCX task, largest on the medium-grained 10-class CIC task, and intermediate on the 20-class USTC task, making CIC-like settings the sweet spot for SSL deployment. In the novel-class setting, SSL maintains high accuracy on unseen attacks, so a new attack can often go online with only 5–50 alerts, reducing latency from days to hours.

This behavior differs from the ImageNet experience of SimCLR and MoCo. On ImageNet, strong augmentation is essential because intra-class visual variation is large. On traffic data, by contrast, discriminative signals are concentrated in high-information dimensions such as packet length, IAT, and TLS-handshake fingerprints, where overly strong perturbations can destroy critical features. Augmentation strength should therefore be adapted to the number of classes: a 2-class task is best left at **none**, a 5–10-class task benefits from **medium**, and a 15+ class task prefers **weak**. The cross-dataset results further show that transfer is not symmetric: a source encoder trained on a harder task can transfer better to an easier target, while embedding-space similarity alone (e.g., $\text{CKA} > 0.99$) is not sufficient to predict transferability.

5.2. SOC deployment, open-set detection, and limitations

In real SOC deployment, the model faces poisoning, adversarial examples, and distribution shift. Poisoning targets SSL pretraining, where injected self-similar pairs can mislead downstream few-shot classification; defenses rely on multi-source trusted data and uniformity statistics of embedding anomalies. Adversarial examples target inference, e.g., small packet-length or IAT perturbations; we leave FGSM/PGD-style evaluation to future work. Distribution shift is the most operationally relevant threat: when a C2 tool changes version and its fingerprint no longer matches, TP-Net’s novel-class few-shot plus continuous micro-fine-tuning can absorb the drift, as five new alerts suffice to rebuild the prototype without retraining the encoder. The resulting security gains are substantial: the retraining window drops from 24–48 hours to hour-level 5-shot labeling, manual annotation drops from thousands of samples to five, and hand-crafted payload signatures no longer need to be maintained. In the open-set setting, the SSL embedding achieves AUROC 0.9127 on CIC and 0.8022 on USTC; known-unknown is easier to detect than true-unknown. For USTC, however, the large number of classes makes the embedding nearly spherical, with a confidence gap of only 0.07, so Mahalanobis or Energy scoring is needed as a supplement.

This work has three main limitations. First, we do not directly compare with large-scale fully supervised SOTA models such as ET-BERT, YaTC, or FS-Net, because their

design objective and corpus scale differ from our frozen-encoder + K -shot episode protocol. Second, the DoHBrw encryption validation and augmentation-strength tuning remain proof-of-concept; more encryption-specific datasets and broader K -shot scans are needed. Third, real SOC deployment requires additional engineering, including Zeek/Suricata event ingestion, ONNX/TensorRT compression, production-grade monitoring and alerting, and a sample-feedback path with existing SIEM systems such as Splunk or ELK. These integration steps are outside the empirical scope of this paper.

6. Conclusion

We have presented TP-Net, a self-supervised pretraining + few-shot evaluation framework for network-traffic novel attack detection, with validation on an encrypted scenario. Its four-branch 1D-CNN encoder fuses packet-level, burst-level, flow-statistical, and handshake features, and is paired with five traffic-specific augmentations. After SimCLR pretraining, the encoder is frozen and evaluated through the N -way K -shot episode protocol; a fine-tune head (`sklearn` LogReg) is further provided as the default SOC deployment option.

Experiments on CIC-IDS2017, USTC-TFC2016, and ISCX-VPN2016 show that TP-Net consistently outperforms random initialization under 5-shot evaluation (+4.11 to +13.14 pp), generalizes to completely unseen attack classes (CIC 2-class novel: 99.3%; USTC 5-class novel: 72.9%), and exhibits asymmetric cross-dataset transfer. The augmentation ablation further reveals that augmentation strength should adapt to the number of classes in the task, rather than simply following the “stronger is better” consensus. On the encrypted-only DoHBrw-2020 subset, SSL pretraining also brings substantial gains, providing preliminary evidence of its effectiveness in pure-encrypted scenarios. Future work includes task-adaptive augmentation scheduling, more refined open-set few-shot detection, cross-dataset transfer analysis, and engineering validation of the fine-tune head paradigm in real SOC deployments.

References

- [1] Cisco Systems, *Cybersecurity Threat Trends: AI, Encryption and the Rise of Encrypted Traffic*, Cisco Cybersecurity Report, 2024.
<https://www.cisco.com/c/en/us/products/security/cybersecurity-reports.html>
- [2] C. Liu, L. He, G. Xiong, Z. Cao, Z. Li, FS-Net: A flow sequence network for encrypted traffic classification, in: *IEEE INFOCOM 2019 - IEEE Conference on Computer*

- Communications*, 2019, pp. 1171–1179.
<https://doi.org/10.1109/INFOCOM.2019.8737507>
- [3] R. Zhao, M. Zhan, X. Deng, Y. Wang, Y. Wang, G. Gui, Z. Xue, Yet another traffic classifier: A masked autoencoder based traffic transformer with multi-level flow representation, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37, No. 4, 2023, pp. 5420–5427.
<https://doi.org/10.1609/aaai.v37i4.25674>
 - [4] X. Lin, G. Xiong, G. Gou, Z. Li, J. Shi, J. Yu, ET-BERT: A contextualized datagram representation with pre-training transformers for encrypted traffic classification, in: *Proceedings of the ACM Web Conference 2022 (WWW '22)*, 2022.
<https://doi.org/10.1145/3485447.3512217>
 - [5] H. Zhang, L. Yu, X. Xiao, Q. Li, F. Mercaldo, X. Luo, Q. Liu, TFE-GNN: A temporal fusion encoder using graph neural networks for fine-grained encrypted traffic classification, in: *Proceedings of the ACM Web Conference 2023 (WWW '23)*, 2023, pp. 2066–2075.
<https://doi.org/10.1145/3543507.3583227>
 - [6] N. Joseph Pramod Dhinakar, K. Ramani, Deep learning approaches to encrypted traffic classification: A comprehensive survey, *AIP Conference Proceedings* 3371 (2026) 030006.
<https://doi.org/10.1063/5.0312319>
 - [7] T. Wang, X. Xie, W. Wang, C. Wang, Y. Zhao, Y. Cui, NetMamba: Efficient network traffic classification via pre-training unidirectional mamba, in: *2024 IEEE 32nd International Conference on Network Protocols (ICNP)*, 2024.
<https://doi.org/10.1109/ICNP61940.2024.10858569>
 - [8] P.H. Le-Khac, G. Healy, A.F. Smeaton, Contrastive representation learning: A framework and review, *IEEE Access* 8 (2020) 193907–193934.
<https://doi.org/10.1109/ACCESS.2020.3031549>
 - [9] L. Jing, Y. Tian, Self-supervised visual feature learning with deep neural networks: A survey, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43 (11) (2021) 4037–4058.
<https://doi.org/10.1109/TPAMI.2020.2992393>

- [10] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, in: *Proceedings of the 37th International Conference on Machine Learning (ICML)*, PMLR 119, 2020, pp. 1597–1607.
<https://proceedings.mlr.press/v119/chen20j.html>
- [11] K. He, H. Fan, Y. Wu, S. Xie, R. Girshick, Momentum contrast for unsupervised visual representation learning, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 9729–9738.
<https://doi.org/10.1109/CVPR42600.2020.00975>
- [12] J.-B. Grill, F. Strub, F. Altché, et al., Bootstrap your own latent: A new approach to self-supervised learning, in: *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33, 2020.
<https://proceedings.neurips.cc/paper/2020/hash/f3ada80d5c4ee70142b17b8192b2958e-Abstract.html>
- [13] X. Chen, K. He, Exploring simple siamese representation learning, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 15750–15758.
<https://doi.org/10.1109/CVPR46437.2021.01549>
- [14] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick, Masked autoencoders are scalable vision learners, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 16000–16009.
<https://doi.org/10.1109/CVPR52688.2022.01553>
- [15] Y. Wang, Q. Yao, J.T. Kwok, L.M. Ni, Generalizing from a few examples: A survey on few-shot learning, *ACM Computing Surveys* 53 (3) (2020) 1–34.
<https://doi.org/10.1145/3386252>
- [16] J. Snell, K. Swersky, R.S. Zemel, Prototypical networks for few-shot learning, in: *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30, 2017, pp. 4078–4088.
<https://proceedings.neurips.cc/paper/2017/hash/cb8da676746899566294d7ebfe6543c8-Abstract.html>
- [17] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, D. Wierstra, Matching networks for one shot learning, in: *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 29, 2016, pp. 3630–3638.

<https://proceedings.neurips.cc/paper/2016/hash/90e1357833654983612fb05e3ec9148c-Abstract.html>

- [18] C. Finn, P. Abbeel, S. Levine, Model-agnostic meta-learning for fast adaptation of deep networks, in: *Proceedings of the 34th International Conference on Machine Learning (ICML)*, PMLR 70, 2017, pp. 1126–1135.
<https://proceedings.mlr.press/v70/finn17a.html>
- [19] F. Sung, Y. Yang, L. Zhang, T. Xiang, P.H.S. Torr, T.M. Hospedales, Learning to compare: Relation network for few-shot learning, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 1199–1208.
<https://doi.org/10.1109/CVPR.2018.00131>
- [20] Z. Hang, Y. Lu, Y. Wang, Y. Xie, Flow-MAE: Leveraging masked autoencoder for accurate, efficient and robust malicious traffic classification, in: *Proceedings of the 26th International Symposium on Research in Attacks, Intrusions and Defenses (RAID '23)*, 2023.
<https://doi.org/10.1145/3607199.3607206>
- [21] M. Rahaman, A. Mahmud, A. Abd Aziz, et al., A systematic literature review on data-efficient and adaptive learning techniques for encrypted traffic classification under modern protocols, *Computers* 15 (5) (2026) 319.
<https://doi.org/10.3390/computers15050319>
- [22] C. Chang, Y. Li, S. Seneviratne, X. Cai, ICT-META: In-context aware few-shot learner for encrypted traffic classification, *IEEE Transactions on Machine Learning in Communications and Networking* 4 (2026) 733–743.
<https://ieeexplore.ieee.org/document/11488357>
- [23] Z. Li, J. Wang, Y.-F. Song, S.-H. Yue, Unlocking few-shot encrypted traffic classification: A contrastive-driven meta-learning approach, *Electronics* 14 (21) (2025) 4245.
<https://doi.org/10.3390/electronics14214245>
- [24] L. Zhao, C. Wu, A. Yu, When pre-training meets contrast learning: Few-shot encrypted traffic classification with novelty detection, *IEEE Transactions on Networking* 34 (2026) 4480–4495.
<https://ieeexplore.ieee.org/document/11435484>
- [25] Y. Qu, C. Zheng, M. Liao, H. Ma, Y. Jiang, W. Wang, SepSpace: Payload-free open-set network traffic classification via supervised contrastive representation learning

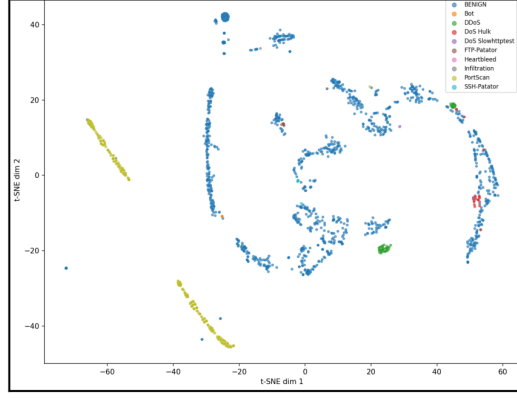
- and prototype-radius rejection, *Cybersecurity* 9 (2026) 204.
<https://doi.org/10.1186/s42400-026-00635-x>
- [26] Y. Yang, Contrastive learning for network intrusion detection: A comprehensive survey, in: *Proceedings of the 2024 2nd International Conference on Computer, Internet of Things and Smart City (CloTSC)*, 2024.
<https://doi.org/10.1145/3731867.3731894>
 - [27] P. Golchin, N. Rafiee, M. Hajizadeh, A. Khalil, R. Kundel, R. Steinmetz, SSCL-IDS: Enhancing generalization of intrusion detection with self-supervised contrastive learning, in: *2024 IFIP Networking Conference (IFIP Networking)*, 2024.
<https://ieeexplore.ieee.org/document/10619725>
 - [28] Y. Jin, J. Fang, Y. Gao, Few-shot encrypted traffic classification model incorporating MAML and contrastive learning, *Journal of Shanghai University (Natural Science Edition)* 31 (4) (2025) 719–734.
<https://www.journal.shu.edu.cn/EN/10.12066/j.issn.1007-2861.2594>
 - [29] Y. Xian, B. Schiele, Z. Akata, Zero-shot learning — the good, the bad and the ugly, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4582–4591.
<https://doi.org/10.1109/CVPR.2017.328>
 - [30] I. Sharafaldin, A.H. Lashkari, A.A. Ghorbani, Toward generating a new intrusion detection dataset and intrusion traffic characterization, in: *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*, 2018, pp. 108–116.
<https://doi.org/10.5220/0006639801080116>
 - [31] W. Wang, M. Zhu, X. Zeng, X. Ye, Y. Sheng, Malware traffic classification using convolutional neural network for representation learning, in: *2017 International Conference on Information Networking (ICOIN)*, 2017, pp. 712–717.
<https://doi.org/10.1109/ICOIN.2017.7899588>
 - [32] G. Draper-Gil, A.H. Lashkari, M.S.I. Mamun, A.A. Ghorbani, Characterization of encrypted and VPN traffic using time-related features, in: *Proceedings of the 2nd International Conference on Information Systems Security and Privacy (ICISSP)*, 2016, pp. 407–414.
<https://doi.org/10.5220/0005740704070414>

- [33] Z. Xu, Deep learning-based intrusion detection systems: A survey, *arXiv preprint arXiv:2504.07839* (2025).
<https://arxiv.org/abs/2504.07839>
- [34] W. Dong, J. Yu, X. Lin, G. Gou, G. Xiong, Deep learning and pre-training technology for encrypted traffic classification: A comprehensive review, *Neurocomputing* 617 (2025) 128444.
<https://doi.org/10.1016/j.neucom.2024.128444>

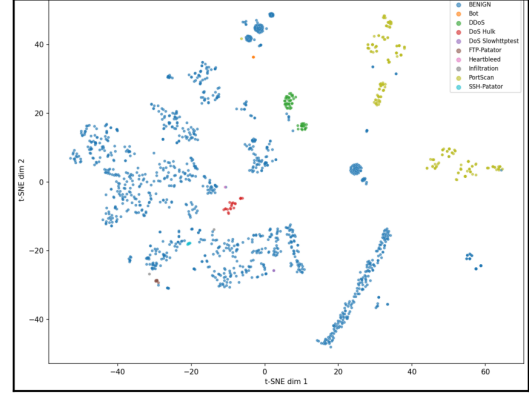
Appendix A. Supplementary experiment tables and visualizations

The supplementary tables and figures cited in the main text are collected below.

CIC-IDS2017

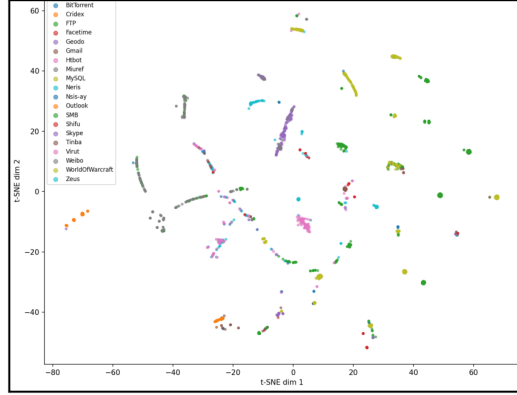


(a1) CIC Random

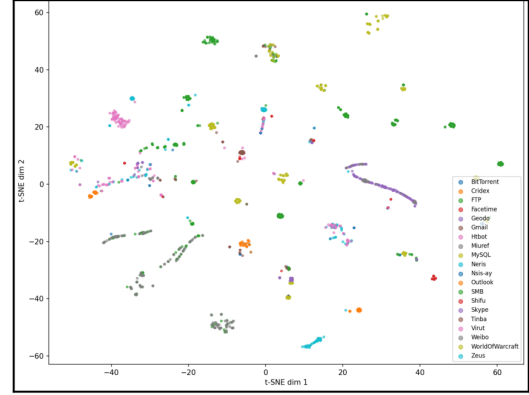


(a2) CIC SSL

USTC-TFC2016

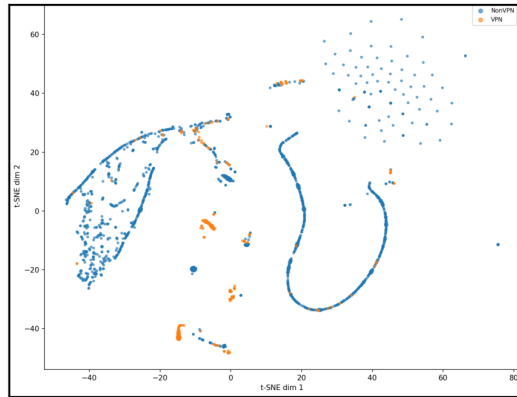


(b1) USTC Random

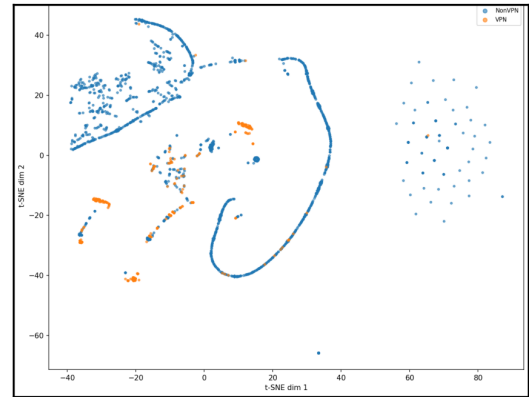


(b2) USTC SSL

ISCX-VPN2016



(c1) ISCX Random



(c2) ISCX SSL

Figure A.1: t-SNE projections (3 datasets $\times$ Random / SSL; subcaptions (a1)–(c2) at the same size as the figure caption).

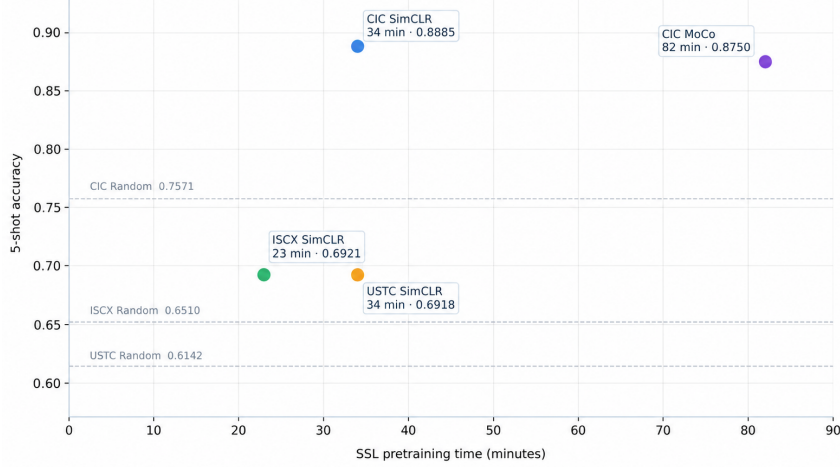


Figure A.2: SSL pretraining cost–benefit scatter.

Table A.1: Full K -shot scan (3 seeds $\times$ 200 episodes; cited in §4.2).

Dataset	K-shot	1	3	5	10
CIC-IDS2017	SSL	0.8521	0.8624	0.8700	0.8752
	Random	0.7728	0.7366	0.7115	0.6799
	Δ (pp)	+7.93	+12.58	+15.85	+19.53
USTC-TFC2016	SSL	0.6621	0.7258	0.7391	0.7525
	Random	0.6995	0.7376	0.7465	0.7498
	Δ (pp)	−3.74	−1.18	−0.74	+0.27
ISCX-VPN2016	SSL	0.6786	0.6407	0.6462	0.6443
	Random	0.6328	0.6266	0.6419	0.6484
	Δ (pp)	+4.58	+1.41	+0.43	−0.41

Table A.2: SSL algorithm diversity (ProtoNet 5-way 5-shot, 3 seeds $\times$ 200 ep, ISCX is 2-way; cited in §4.3).

SSL algorithm	Paradigm	CIC-IDS2017	USTC-TFC2016	ISCX-VPN2016	3-dataset mean
SimCLR	Contrastive (in-batch negatives)	0.8700 $\pm$ 0.004	0.7391 $\pm$ 0.008	0.7026 $\pm$ 0.008	0.7706
MoCo	Contrastive (momentum queue)	0.8537 $\pm$ 0.002	0.6606 $\pm$ 0.006	0.6249 $\pm$ 0.012	0.7131
BYOL	Bootstrapping (EMA target)	0.8676 $\pm$ 0.001	0.6802 $\pm$ 0.004	0.6104 $\pm$ 0.010	0.7194
SimSiam	Bootstrapping (stop-gradient)	0.8465 $\pm$ 0.003	0.6592 $\pm$ 0.008	0.5573 $\pm$ 0.013	0.6877
Embedding-MAE†	Reconstructive (embedding-dim mask)	0.7886 $\pm$ 0.003	0.6121 $\pm$ 0.005	0.5806 $\pm$ 0.011	0.6604

† Embedding-MAE (simplified version in this paper) $\neq$ the MAE of He et al. 2022; see §4.3.

Table A.3: Extended SimCLR heads comparison (MoCo / MatchingNet columns removed from Table 3 for compactness; full data restored here, 5 seeds $\times$ 600 episodes).

Dataset	Task	SimCLR + MoCo	SimCLR + MatchingNet	$\Delta_{\text{Proto-Match}}$ (pp)
CIC-IDS2017	5-way 5-shot	0.8750 ± 0.003	0.8760 ± 0.003	+1.25
USTC-TFC2016	5-way 5-shot	0.5880 ± 0.005	0.6203 ± 0.005	+7.15
ISCX-VPN2016	2-way 5-shot	0.6223 ± 0.003	0.6097 ± 0.003	+8.24

Note: $\Delta_{\text{Proto-Match}} = (\text{SimCLR} + \text{ProtoNet}) - (\text{SimCLR} + \text{MatchingNet})$, i.e. the metric-learning gain over cosine attention. *SimCLR* > *MoCo* by +1.35 pp on CIC (0.8885 vs 0.8750); *MoCo* is worse than Random on USTC/ISCX.

Table A.4: USTC per-class 5-shot accuracy (SSL vs Random, 5 seeds $\times$ 600 episodes; top 5 + bottom 5, 10 classes; cited in §4.5).

Application class	SSL	Random	Δ (pp)
— SSL top 5 (SSL strongest) —			
Outlook	0.9906	0.9920	−0.14
WorldOfWarcraft	0.9822	0.9999	−1.77
Tinba	0.9280	0.3932	+ 53.48
Skype	0.9021	0.9142	−1.21
Facetime	0.8822	0.8891	−0.69
— SSL bottom 5 (SSL weakest) —			
BitTorrent	0.5196	0.5845	−6.49
Neris	0.5153	0.3855	+ 12.98
Gmail	0.5120	0.5164	−0.44
Miuref	0.5024	0.3135	+ 18.89
Nsis-ay	0.5007	0.2413	+ 25.94

Table A.5: Open-set FSL evaluation (CIC / USTC, 3 seeds $\times$ 200 ep; cited in §5.2).

Dataset	Scenario A closed-set acc	Scenario B AUROC	Scenario C AUROC	Scenario B conf gap	Scenario C conf gap
CIC-IDS2017	0.8261	0.9127 $\pm$ 0.004	0.8285 ± 0.002	0.51	0.29
USTC-TFC2016	0.8736	0.8022 ± 0.003	0.7202 ± 0.008	0.07	0.07

Data availability

The three public traffic datasets used in this work are openly available: CIC-IDS2017 (<https://www.unb.ca/cic/datasets/ids-2017.html>), USTC-TFC2016 (<https://github.com/yungshenglu/USTC-TFC2016>), and ISCX-VPN2016 (<https://www.unb.ca/cic/datasets/vpn.html>). The DoHBrw-2020 dataset is publicly distributed by the Stratosphere IPS Lab (<https://stratosphereips.org/category/dataset.html>). All preprocessing scripts, pretrained checkpoints, and per-episode results will be released at an anonymous GitHub

repository (<https://github.com/YUN625-art/TP-Net>, link to be replaced with the final URL upon acceptance).